S1Q12

Question 12 - Stage 1

The parameters a, b and c each take the value 0 or 1.

There are eight such triples to consider.

Find (to 3 d.p.) the value of the expression $\sum_{a,b,c} \int_{-1}^{c} (x-a)^b (x+b)^c x^a \, \mathrm{d}x$

Working

In [1]:
```
import scipy.integrate as integrate
from itertools import product

sumTotal = 0
error = 0

for combination in product([0,1],repeat=3):
    a,b,c = combination
    result = integrate.quad(lambda x: ((x-a)**b)*((x+b)**c)*(x**a),-1,c)
    sumTotal += result[0]
    error += result[1]

print(f"Error: {error}")
print(f"Total: {round(sumTotal,3)}")
```

```
Error: 6.648911753050776e-14
Total: 2.167
```